

Sound studio

Revise 360 lesson plan — 60 minutes

Unit	OCR J277 1.2 Memory and storage — Paper 1, Computer systems
Lesson	Sound studio (lesson 14 of the unit)
Lesson ID	ms-l14
Experience ID	ms-l14
Plan ID	PLAN-MS-L14
Content version	f3d204df8751
Website	https://revise360.co.uk/experience.html?id=ms-l14

Big question, outcomes and success criteria

Big question — How do computers store sound, and what are the implications for sample rate, duration and bit depth?

Prerequisites

The lesson before this one is ms-l13. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: ms-l01, ms-l02, ms-l03, ms-l04, ms-l05, ms-l06, ms-l07, ms-l08, ms-l09-test, ms-l10, ms-l11, ms-l12.

Assessable outcomes

1. **ms-l14-o1** — understand how sound is sampled and stored in digital form
2. **ms-l14-o2** — understand how sample rate, duration and bit depth affect file size and quality

Vocabulary, resources and preparation

Key vocabulary — Sampling, Sample rate

Resources

SLIDES	presentations/MS_L14_SoundStudio_Lesson.pptx
WORKSHEET	worksheets/MS_L14_SoundStudio_Worksheet.docx
ONLINE EXPERIENCE	experience.html?id=ms-l14

Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.

2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

Worth knowing before you promise anything. In this build APP_CONFIG.backendUrl is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Retrieve	Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites.	Answer the starter on the worksheet.	The starter box on MS_L14_SoundStudio_Worksheet.docx is filled in.
5-8	Review the starter	Take answers; correct the one most got wrong.	Correct their own starter.	A correction written, not just a tick.
8-18	Explain and model	The new idea in two or three steps, with the example worked in front of the class.	Listen and answer questions about the example.	They can say why, not only what.
18-21	Set up and settle	Sign-in, launch, and what to write before anyone taps an answer.	Sign in, open the lesson, find Analogue to digital.	Everyone is in and on station 1.
21-43	Online: work the stations in order	Pupils read each wall, write the key fact and the challenge, then answer at the badge.	Read each wall, write the key fact and the challenge, then answer at the badge.	The worksheet has writing on it for each of the 6 stations before the badge is tapped.
43-48	Check understanding	The hinge question, hands down.	Answer the hinge question, hands down.	The split across the options.
48-56	Apply	Exam practice on the worksheet, in silence.	Exam practice, in silence.	worksheet, Exam practice a) to c)
56-60	Exit and set up the next lesson	The exit question and the retrieval link to next lesson.	Write the exit answer.	One sentence that answers the big question.

The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

1. Analogue to digital

- Sound is an analogue wave: it changes smoothly and continuously.
- Computers store discrete numbers, so the wave must be measured at intervals.
- Each measurement is called a sample, and is stored in binary.

Challenge on this wall: Why can't a computer store a sound wave exactly?

2. Sample rate

- Sample rate is the number of samples taken per second, measured in hertz.
- A higher rate follows the wave more closely, so the recording is more accurate.

- CD quality is 44,100 Hz.

Challenge on this wall: What happens to the recording if the sample rate is too low?

3. Bit depth

- Bit depth is the number of bits used to store each sample.
- More bits means more possible levels, so the amplitude is recorded more precisely.
- 3 bits give 8 levels; 16 bits give 65,536.

Challenge on this wall: How many levels does a bit depth of 4 give?

4. Sound file size

- File size (bits) = sample rate × bit depth × duration in seconds.
- Double any one of them and the file size doubles.
- Stereo doubles it again, because each channel is stored separately.

Challenge on this wall: Calculate the size of 10 seconds at 8,000 Hz with 8-bit samples.

5. Quality against size

- Higher sample rate and bit depth mean better quality and bigger files.
- Speech needs far less than music, so phone calls use low rates.
- Choose what's good enough for the purpose.

Challenge on this wall: Why is a phone call sampled at a much lower rate than a music track?

6. In the studio

- Practise reading a wave and choosing the nearest level at each sample.
- Then calculate the file size from your choices.
- This is exactly what an exam question on sound representation looks like.

Challenge on this wall: Complete both sampling tasks on your worksheet.

Retrieval and the check for understanding

Starter — Sound is a smooth wave, but computers store numbers. Suggest how a recording could be turned into numbers, and what is lost.

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (ms-114-s4-c2), so the wording a pupil sees matches the wording you say.

Which change would halve a sound file's size?

Correct: Halving the sample rate

What a pupil might choose instead

- Doubling the bit depth
- Recording in stereo
- Adding metadata

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

The course's own response to a wrong answer here: Half as many samples, half the size, and some loss of quality.

One response, not four. one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
Recognise	11	The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.
Construct	4	Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer.

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

The stations, and what to record at each

#	STATION	ACTIVITIES	RECORD ON PAPER
1	Analogue to digital	2	Sampling means...
2	Sample rate	2	Sample rate is...
3	Bit depth	2	Bit depth is...
4	Sound file size	2	Sound file size is calculated by...
5	Quality against size	2	Quality and file size are linked because...
6	In the studio	2	To sample a wave, I should...
final	Final challenge: sample and calculate	3	the answer to the station's challenge

Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

Assessment, evidence and unfinished work

ID	QUESTION	MARKS	WHERE
ms-114-x1	Describe what happens when a sound wave is sampled.	2	worksheet, Exam practice a); deck, Exam practice slide
ms-114-x2	Explain the effect on quality and file size of doubling the bit depth.	3	worksheet, Exam practice b); deck, Exam practice slide
ms-114-x3	A recording lasts 45 seconds at 12,000 Hz with a bit depth of 16. Calculate the file size in kilobytes, showing your working.	4	worksheet, Exam practice c); deck, Exam practice slide

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/ms-114.json`, which is not published with the website.

Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **What is sampling?** — Playing a sound back at a lower speed; Copying part of one song into another
- **What does sample rate measure?** — How many bits are used for each sample; How long the recording lasts
- **What is the effect of doubling the sample rate?** — Better quality with no change in size; Worse quality and a smaller file
- **What is the effect of increasing the bit depth?** — More samples are taken each second; The recording lasts longer
- **A 30-second clip is sampled at 44,100 Hz with a bit depth of 16. What — 1,323,000; 705,600**

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

One of these, written, not a show of hands:

- How sampling works
- Sample rate and bit depth
- Calculating sound file size

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

Retrieval link to next lesson

Next is ms-115, Compression works. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

Shorter lessons, longer lessons and no internet

A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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